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The *Wasp* Sampling Policy for High-Speed Network Intrusion Detection Systems

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Introduction



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- Cloud and edge infrastructures **operate** at **multi-gigabit** speeds demanding real-time intrusion detection
- Modern NIDS rely on **machine learning** techniques for anomaly detection
- The **challenge** is scaling to wire speeds without compromising detection accuracy or requiring specialized hardware

State of the Art



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- **eBPF** processes packets immediately upon NIC arrival
- Safety checks performed by the **verifier** to ensure programs cannot compromise the kernel
- Verified bytecode is **JIT compiled** into native machine code for optimal runtime performance
- The **loader** injects eBPF programs into the kernel dynamically without requiring its recompilation

State of the Art



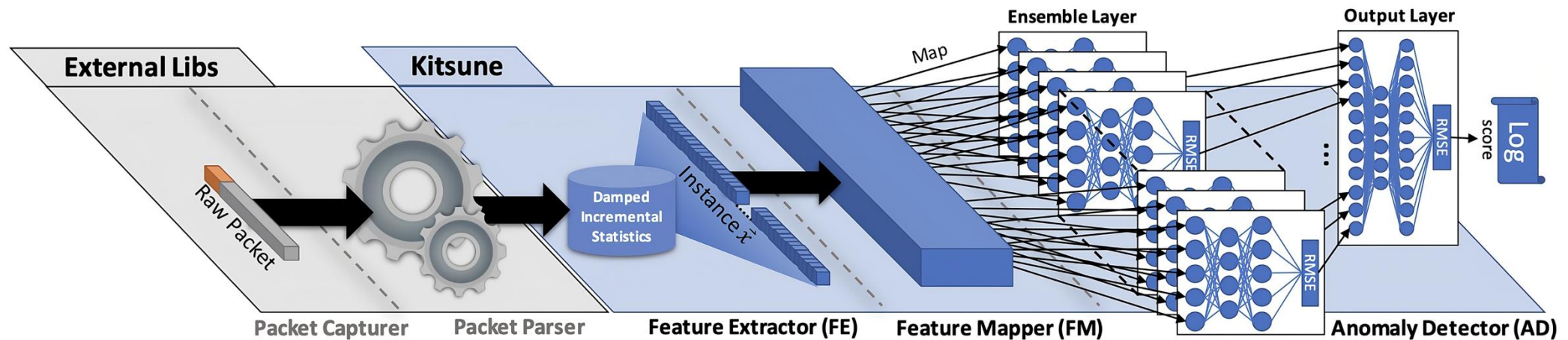
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- **Kitsune** is one of the most popular NIDS in the research community
- A packet is **classified** as malicious when its reconstruction RMSE exceeds the learned anomaly threshold

Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection by Mirsky et al.



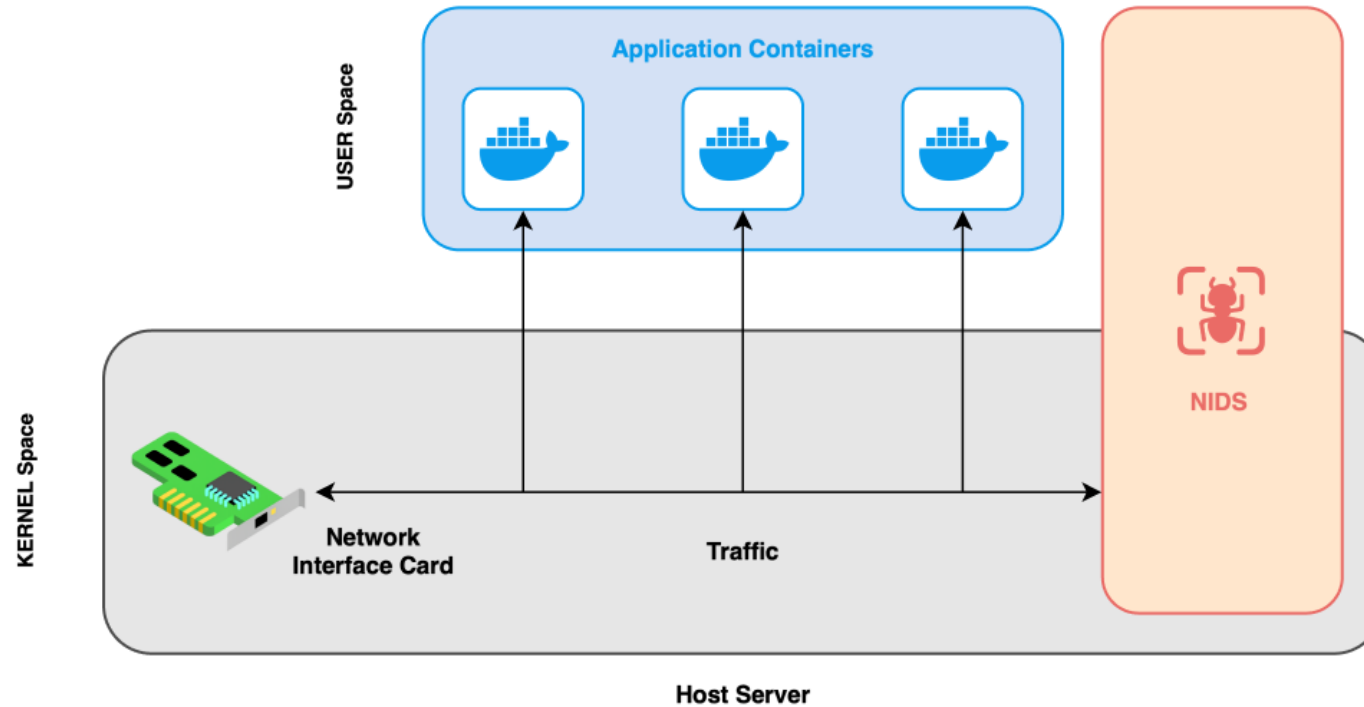
Deployment Scenario



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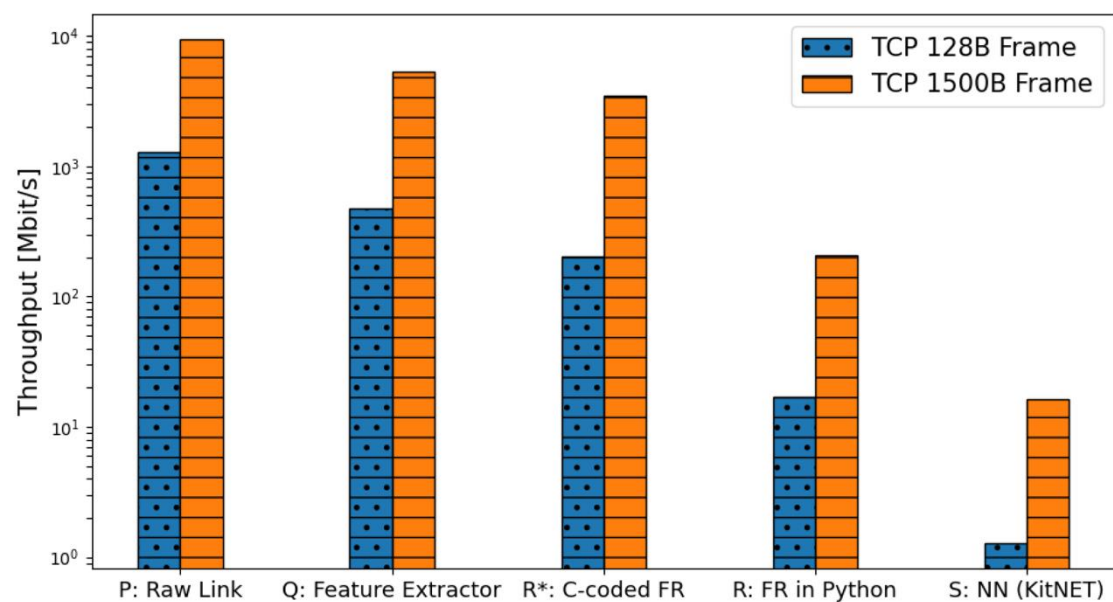
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Throughput



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Protectli VP6650 equipped with 10 Gbit/s interfaces



Contributions



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- Implemented eBPF-based **feature extraction** in kernel space to achieve multi-gigabit throughput
- Designed **Wasp**: a sampling policy that addresses the inference bottleneck at wire speeds
- Introduced **flow-aware** packet classification to assign labels of unsampled traffic instead of random guessing

- **Uniform** sampling with p calibrated to match ring buffer capacity

$$X_i \sim \text{Bernoulli}(p)$$

- **Flow-aware** prediction labels all packets in a flow as malicious once the flow is flagged
- The core **assumption** is that network attacks involve packet sequences and not isolated events

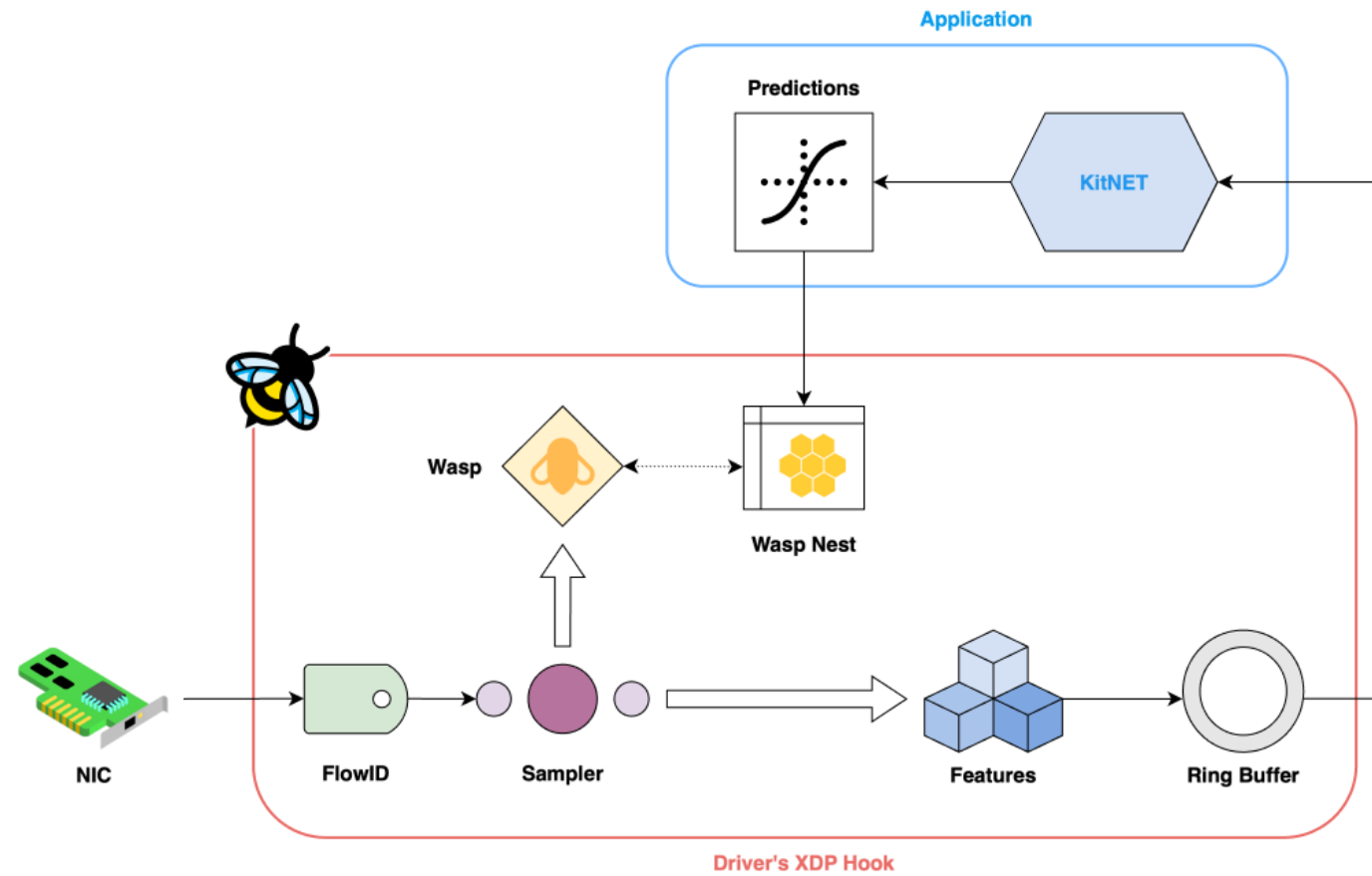
Wasp



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Testing



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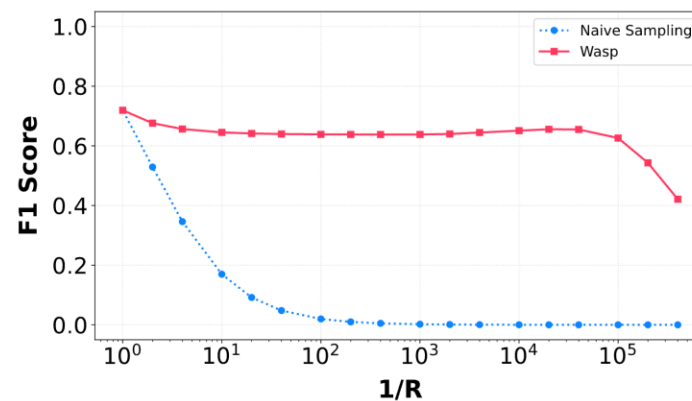
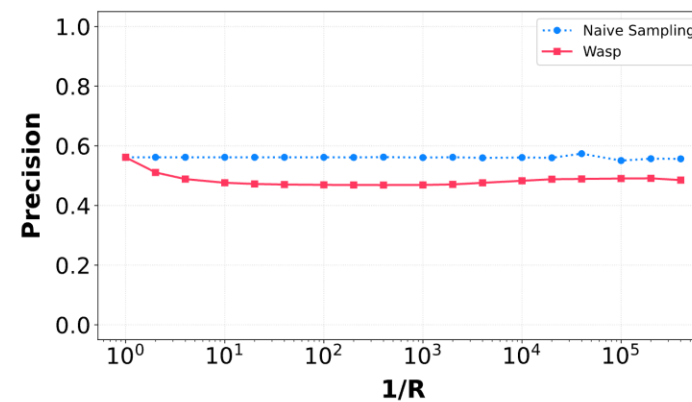
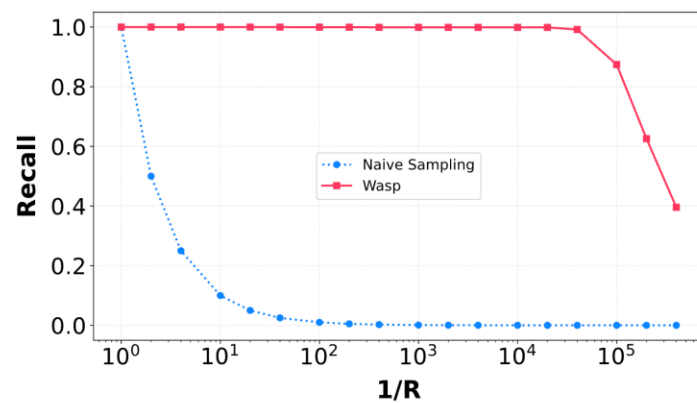
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- **Evaluation** performed on Kitsune Network Attack Dataset and TON-IoT
- Naïve sampling serves as **baseline** where unsampled packets are labeled as benign simulating missed detections
- KitNET **trained** on ~55.000 benign packets to learn legitimate traffic behavior and establish the anomaly threshold
- Performance evaluated through confusion matrix **metrics** such as F1-score, recall, and precision

ARP Man in the Middle



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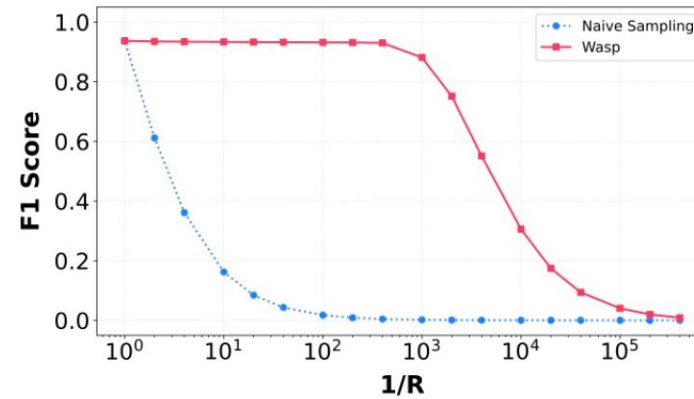
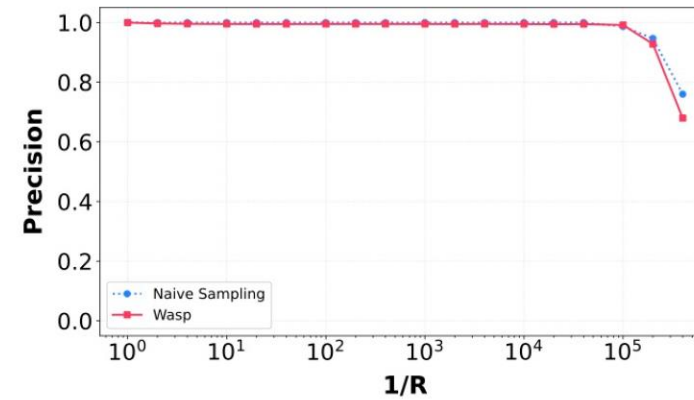
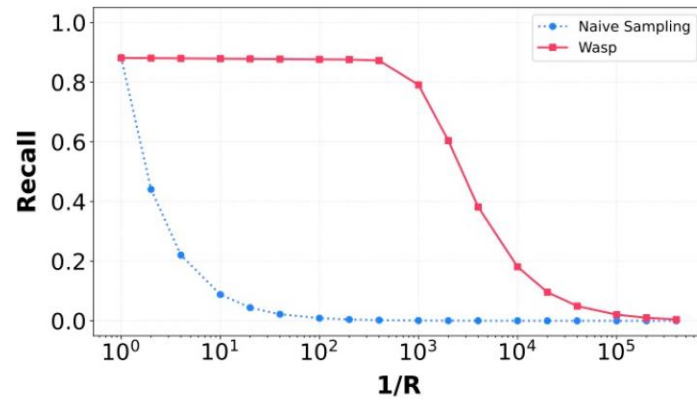
Mirai Botnet



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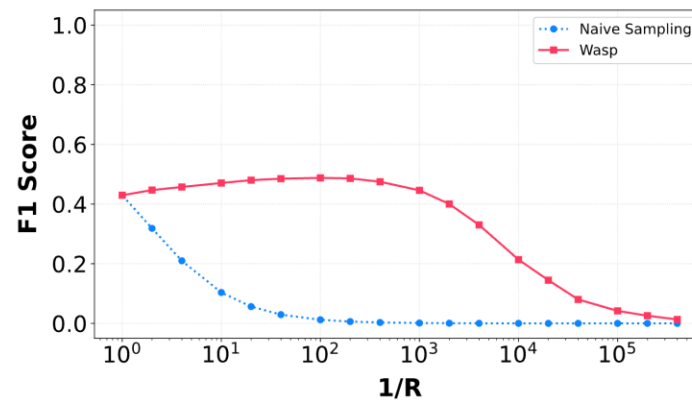
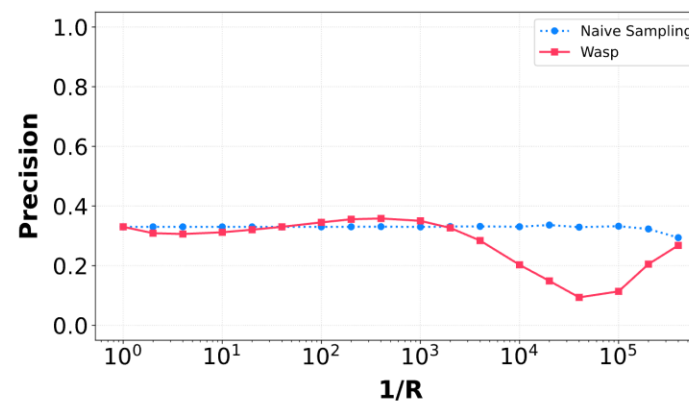
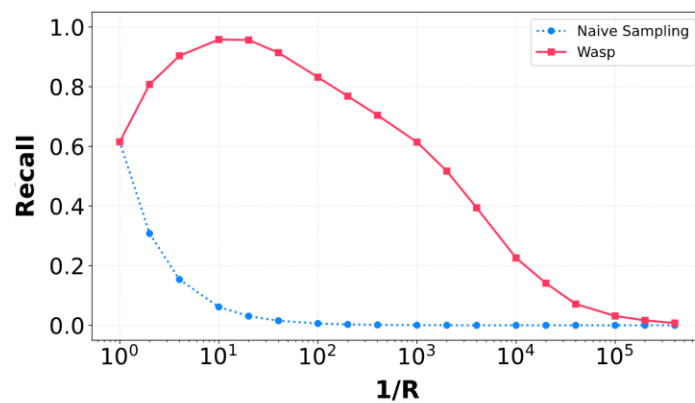
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Ransomware



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Conclusions



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- Kernel-level feature extraction **enables multi-gigabit** packet processing
- Sampling is **key** to implementing anomaly detection at wire speed as user space cannot keep pace
- Wasp **preserves** detection effectiveness at sampling rates down to $\sim 1/10^3$ for most attack scenarios
- **Flow-aware** labeling outperforms stateless approaches like Naïve Sampling

An Investigation on Packet Sampling between Kernel and User Space for NIDS

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Abstract— Extended Berkeley Packet Filter technology has been successfully used to accelerate several data-plane algorithms. An application area of growing interest is Intrusion Detection, where timely packet processing at high speed is critical. In this paper, we focus on anomaly detection, which uses machine learning to identify packets belonging to a malicious flow with the intervention of a packet sampling policy to keep up with the traffic network pace in a kernel-to-user-space pipeline, to investigate the deployment feasibility of the designed anomaly-based kernel-enhanced intrusion detection system. The performance tests related to the packet sampling policy have been carried out taking into account the same dataset used to test the inference algorithm, establishing a packet sampling rate threshold maintaining a high accuracy. The throughput measurements have been tried out on our testbed composed by two back-to-back connected programmable middlebox leveraging on the iperf3 tool to employ the stress test, validating that our designed network intrusion detection system is suitable for deployment.¹

Index Terms— packet sampling policy, ebpf, packet sampling rate, feature extraction, nids, online learning

In this paper, we present a software NIDS based on Mirsky's Kitsune [1] that can be deployed either in a programmable middlebox, or in a compute node in a data center. In both scenarios we only consider boxes equipped with general purpose processing hardware even though we expect that boxes equipped with GPUs or AI accelerators will be increasingly common in the near future.

Our NIDS comprises three major components: the feature extractor, the sampler, and the anomaly detection. The feature extractor is an optimized version of Mirsky's feature extractor re-implemented using extended Berkeley Packet Filter (eBPF) [2]. The new feature extractor can process packets at multi-gigabit speed on a general purpose CPU. The anomaly detector is the same as Mirsky [1], which processed packets at a much lower rate. In order to match the rates of the two components we introduce a sampler, which sends to the anomaly detector only a subset of the packets. All other packets are labeled with a much faster algorithm, which labels a packet as malicious if



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Thank You
Questions?

Policies



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